Moving from Reactive to Proactive Retention

StreamMax

The Business Challenge

- Traditional OTT churn systems identify users **after cancellation** behavior begins, causing expensive last-minute campaigns.
- StreamMax aims to detect **Engagement Fatigue** — the gradual decline in viewing momentum that occurs before churn.

Analytical Objective

- Predict probability of engagement fatigue
- Rank users by disengagement risk
- Enable early, cost-efficient intervention

Our Framework — Decode. Decide. Deliver.

STEP 01

Decode

Analyze behavioral signals such as viewing minutes, session frequency, and inactivity patterns.

STEP 02

Decide

Engineer momentum-based features capturing usage decay and habit disruption.

STEP 03

Deliver

Deploy an XGBoost-based fatigue scoring engine powering retention decisions.

Understanding the Signals Behind Engagement Fatigue

01 Engagement Momentum > Absolute Usage

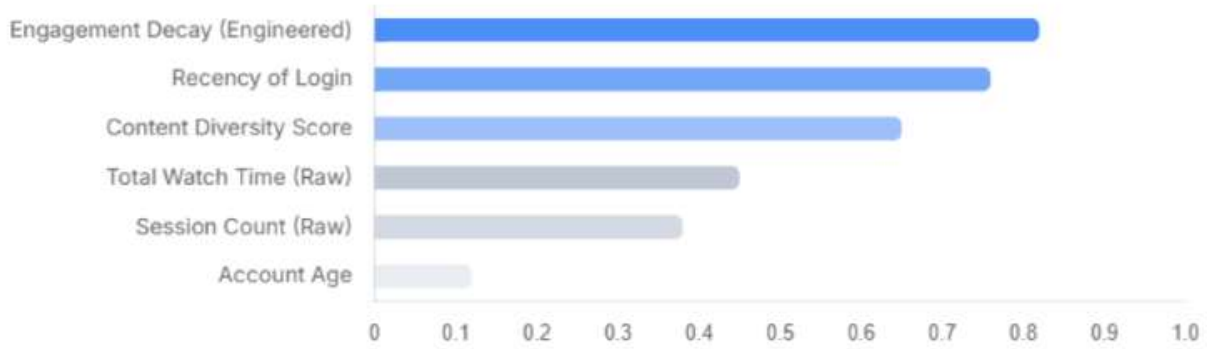
Absolute watch time is not predictive alone. The strongest fatigue indicator captures the rate of change:

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Engagement Decay = Recent Usage ÷ Historical Usage
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↘ Users whose 7-day engagement falls below normal behavior show early fatigue signals.

FEATURE PREDICTIVE POWER (CORRELATION)

Engineered vs. Raw



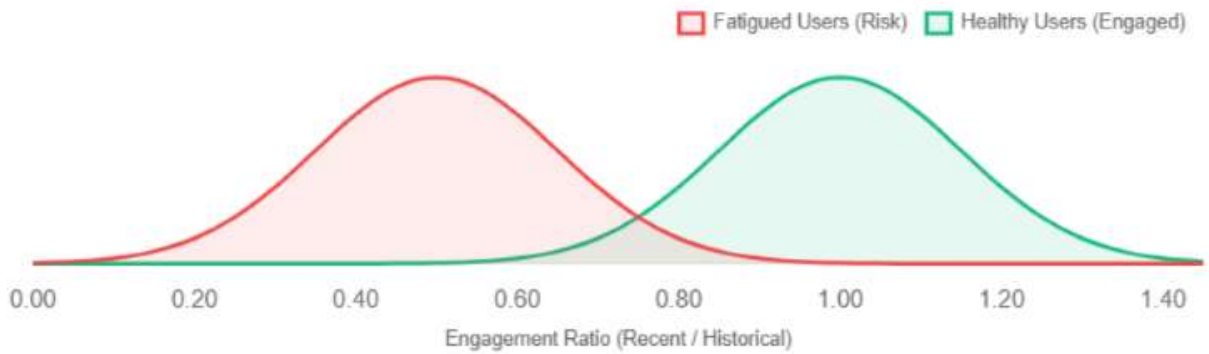
02 The Ghosting Effect

Silence speaks volumes. Inactivity duration is a critical predictor of imminent churn.

3x Likelihood to transition into fatigue if inactive for >7 days compared to active users.

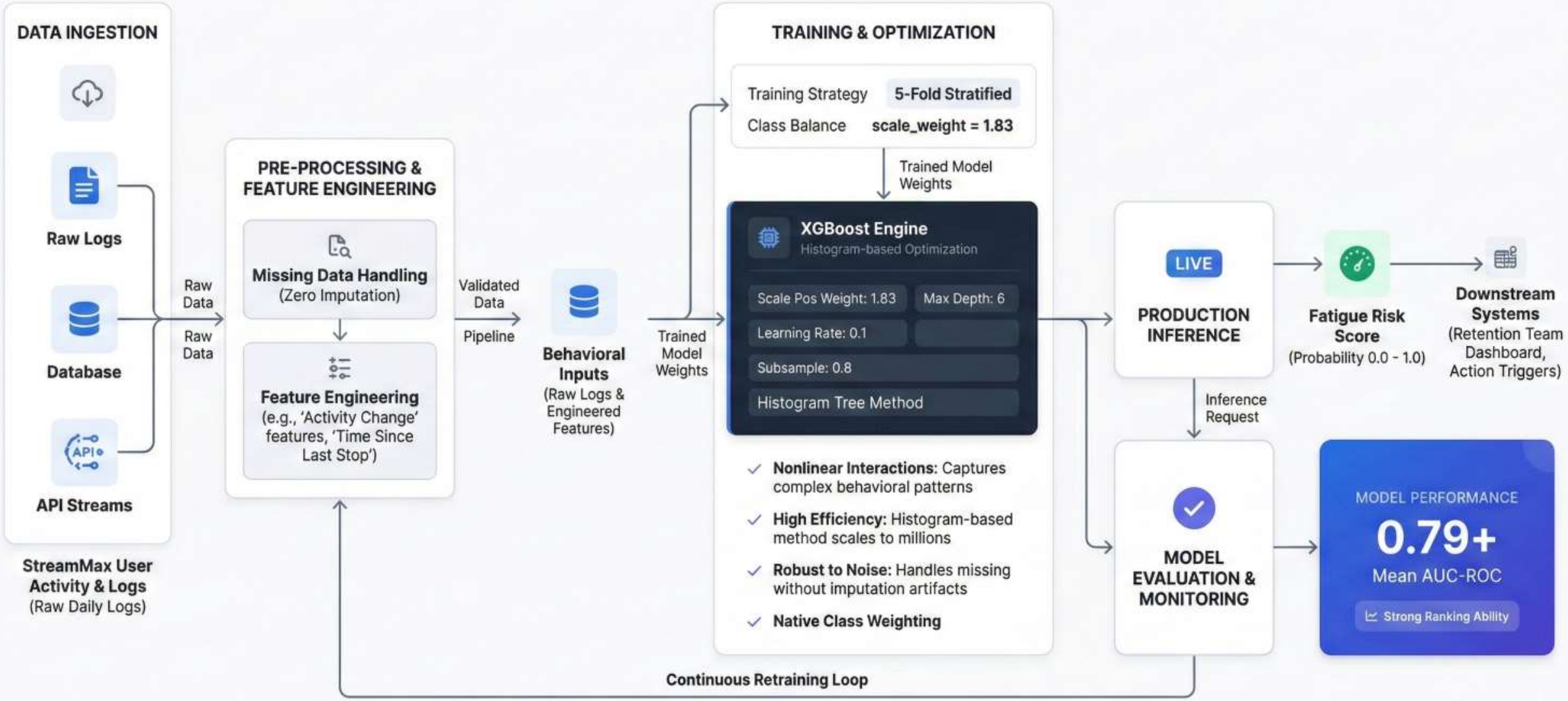
ENGAGEMENT DECAY: ENGAGED VS. FATIGUED

Distribution Analysis

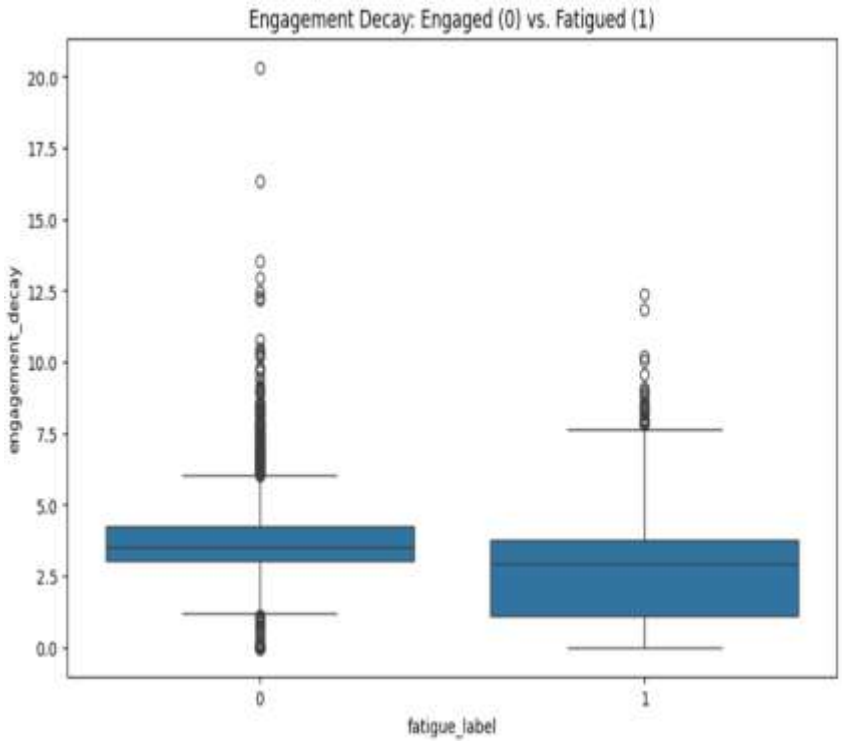


Deciding on the Model — Precision Risk Ranking

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Key Drivers of User Disengagement



- **Business interpretation:** User churn is rarely sudden., It begins with behavioral momentum loss.
- Our model identifies Engagement Fatigue,7–14 days before cancellation risk.
- the thing is the 90% issue is not about the content but its depending on the psychological and customer behaviour even we get bored after watching for sometime that is the main thing is how much the user is getting satisfied so involved the feature engagement decay and it is the high valued and has more importance .
- To solve the issue we need to involve risk segmentation logic

Risk Level	Behavior Pattern	Business Meaning
Low	Stable engagement	Healthy subscriber
Medium	Rising rec_failure	Discovery frustration
High	Engagement decay	Habit breaking
Critical	Ghosting	Imminent churn

- **The "Vanishing Veteran" (High satisfaction_decay):** Long-tenure users who are finishing less content. They aren't bored; they are disappointed.
- **The "Search-Fatigued" User (High rec_failure):** Users who search heavily but click rarely. Their discovery loop is broken.
- **The "Ghosting" Subscriber (High is_ghosting):** Users who have broken their daily habit. Every day they stay away, their probability of return drops by ~12%.

The Tiered Intervention Matrix — Maximizing ROI

✅ **Executive Insight.** StreamMax shifts from reactive discounting to AI-driven precision retention.

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Intervention Strategy Matrix

RESOURCE ALLOCATION

RISK SEGMENT	FATIGUE PROB. (P)	PRIMARY DRIVER	TARGETED INTERVENTION	COST TO STREAMMAX
Healthy	< 0.40	Engaged	**Organic Experience:** No action required.	ZERO
Mild Risk	0.40 - 0.65	rec_failure	**"Content Refresh":** UI switch to human-curated "Staff Picks" or "New Season" alerts.	Near-Zero
High Risk	0.65 - 0.85	eng_decay	**"Surprise & Delight":** 7-day Premium Tier trial or early access to an "Original" pilot episode. [cite: prompt]	Low
Critical	> 0.85	is_ghosting	**"Financial Hook":** Proactive 3-month "Loyalty Pricing" (e.g., 50% off) to prevent total churn.	High

$$ROI = (P_{fatigue} \times \text{Retention Lift} \times ARPU) - \text{Intervention Cost}$$

Delivering Proactive Retention at Scale

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SYSTEM ARCHITECTURE

Data Pipeline
Snowflake / Spark ETL

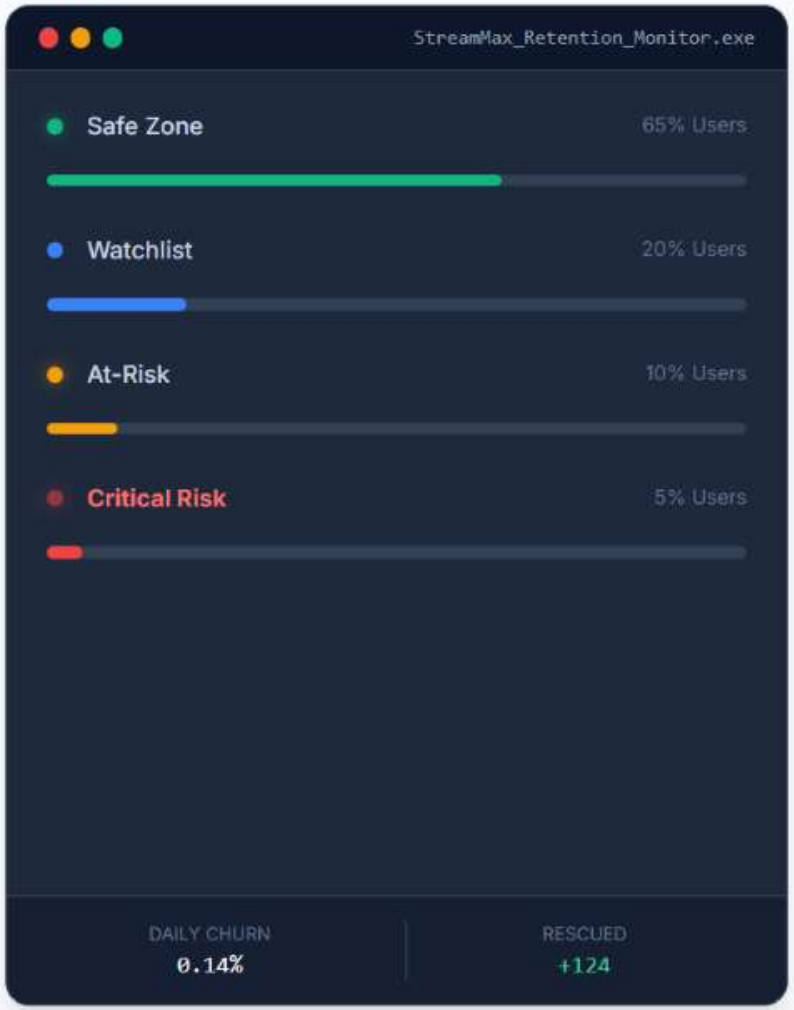
Scoring Engine
XGBoost (Batch: 24h)

CRM Integration
Automated Triggers

SCALABILITY ADVANTAGE

- Milliseconds/User
- CPU Optimized

OPERATIONAL DASHBOARD



BUSINESS IMPACT

Annual Revenue Recovery

$480,000

CALCULATION LOGIC

- Users: 1M
- Churn: 5%
- **Saved: 10%**
- ARPU: $8

FUTURE ENHANCEMENTS

- **Content Awareness:** Distinct models for Sports vs. Movies behavior.
- **Real-time Adaptation:** Adjust recommendations immediately during session.
- **Feedback Loop:** Integrate customer survey data for sentiment analysis.